PROJECT REPORT

Global Sales Dashboard using Power BI

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1. Introduction

Data visualization plays a crucial role in understanding business performance and making informed decisions. Power BI is a powerful business intelligence tool used to transform raw data into meaningful insights.

In this project, a Global Sales Dashboard is developed using Microsoft Power BI to analyze sales performance across different regions, markets, and customer segments.

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2. Problem Statement

To design an interactive dashboard that:

- Provides a clear overview of global sales performance
- Identifies top-performing and underperforming products
- Analyzes sales distribution across regions and markets
- Helps businesses make data-driven decisions

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3. Dataset Description

The dataset contains global sales data with the following attributes:

- Sales Amount
- Quantity Sold
- Delivery Time
- Return Orders
- Customer Segment
- Market (Asia Pacific, Europe, USCA, LATAM)
- Product Details
- Region (Geographical Data)

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4. Tools & Technologies

- Microsoft Power BI
- Data Modeling
- DAX (Data Analysis Expressions)
- Data Visualization Techniques

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5. Dashboard Overview

The dashboard consists of multiple interactive components:

KPIs (Key Performance Indicators)

- **Total Sales:** $12.64M
- **Total Quantity:** 178K
- **Average Delivery Days:** 4
- **Return Orders:** 1079

Visualizations Used

- Sales by Segment (Pie Chart)
- Sales by Market (Donut Chart)
- Sales by Region (Map Visualization)
- Top 10 Customers (Bar Chart)
- Top 6 Profit Products (Bar Chart)
- Top 6 Loss Products (Bar Chart)

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6. System Architecture

Data Source → Data Cleaning → Data Modeling → DAX Measures → Visualization → Dashboard

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7. Methodology

Step 1: Data Import

Data is imported into Power BI from source files.

Step 2: Data Cleaning

- Handling missing values

- Formatting columns
- Removing inconsistencies

Step 3: Data Modeling

- Relationships created between tables
- Data structured for analysis

Step 4: DAX Calculations

Key measures created:

- Total Sales
- Total Quantity
- Average Delivery Time
- Return Orders

Step 5: Visualization

- Charts and graphs created for insights
- Map used for geographical representation

Step 6: Dashboard Design

- Interactive layout with filters (Year: 2012–2015)
- User-friendly UI

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8. Results & Insights

Key Insights:

- Consumer segment contributes the highest sales
- Asia Pacific and Europe are major revenue-generating markets
- Some products generate high profit while others incur losses
- Top customers significantly contribute to overall revenue
- Return orders indicate areas for improvement

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9. Advantages

- Easy data interpretation through visuals
- Interactive and user-friendly dashboard
- Helps in business decision-making
- Real-time insights capability

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10. Limitations

- Depends on data accuracy
- Static dataset (not real-time)
- Limited predictive analysis

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11. Applications

- Business performance monitoring
- Sales strategy optimization
- Customer analysis
- Market trend analysis

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12. Future Scope

- Add predictive analytics
- Integrate real-time data sources
- Deploy dashboard on cloud (Power BI Service)
- Add more advanced KPIs

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13. Conclusion

The Global Sales Dashboard successfully visualizes sales data and provides meaningful insights into business performance. Using Power BI, complex data is transformed into an interactive dashboard that helps in strategic decision-making and performance tracking.